

Project Y

AP Physics 2 · nuclear

A mesa laboratory in 1943. Chalkboards and typed sheets, and no screens anywhere.

49	8	5.5	8/10	3	11 · 36 · 2
STOPS	CHOICE (16%)	EFFECTIVE FORMATS	EQUATIONS COMPUTED	SELECT-ONLY CONCEPTS	SELECT · CONSTRUCT · OPERATE

READING Where this campaign stands

Healthy: 8 of 10 computed, and the one true absence is $A = \lambda N$ — activity from a decay constant and a population, mentioned nowhere. The other gap sits on a TRACE and is probably a Phase 0 phantom. The constraint that shapes everything here is the setting: 1943, chalkboards and typed sheets, **no screens anywhere**. That rules out any panel that looks like a monitor and it rules in the instruments of the period, which happen to be the most tactile ones in the registry.

NOW The mix as it ships



CURRICULUM What the syllabus is not getting

EQUATION THE COURSE OWES	WHAT IT CARRIES	MENTIONED AT	STATUS
$A = \lambda N$	activity from a decay constant and a population	– no stop at all	real gap
$I \propto 1 / r^2$	inverse-square fall-off of radiation intensity	14 · TRACE	verify first

1 of 2 sit on an instrument stop, whose board arithmetic the coverage check cannot yet read — expect Phase 0 to clear those. 1 is unaffected by it.

TIER Concepts the player only ever picks from a list

#	MECHANISM	CONCEPT	STOPS
9	Fusion		41, 46
15	Conservation of energy and momentum in reactions		32
21	Isotope separation and enrichment		43

Reported, never failed. Select-tier can be exactly right; these are the candidates, not the defects.

FUN What this subject and this place can carry

TALLY

STRONG

a T or P stop

A scaler counting clicks is the defining instrument of 1943 physics, and TALLY is that instrument: take counts in batches, watch the rate firm up, and decide when there is enough to report. It also computes the one equation the game genuinely does not have.

SERVES $A = \lambda N$ — the only true absence on the syllabus

NEEDS an early batch that flatters the number, and Poisson scatter authored honestly

HOLD STRONG

P · Experimental Physics

Hold pile power inside a closing band on control rods. This is not an analogy — it is what the period’s reactor operators did, with a needle and no computer, and a do-nothing run kills the experiment.

SERVES criticality and multiplication as a controlled rate

NEEDS the band closing from the physics, not from a difficulty curve

PROBE STRONG

a P stop

Walk a counter out from a source and take readings station by station. Inverse square is the one law you can discover with a trolley and a chalk mark, and the game has an existing TRACE about whether the neutrons or the counts went missing.

SERVES $I \propto 1/r^2$, and it needs no screen at all

NEEDS a break that is not the worst number

SWEEP STRONG

a T or X stop

Reflector thickness against critical mass, on a plot you build by looking. Period-true: a galvanometer and graph paper, one control at a time.

SERVES concept 15 and the tamper argument

BELT MEDIUM

CM · Chemistry & Metallurgy

A sorting bench in metallurgy, one bit of subject matter at speed. Honest, but a calutron is not a conveyor, and concept 21 — isotope separation — deserves a format that teaches the separation rather than a sort.

SERVES concept 21, weakly

TRIAL MEDIUM

the mesa

A real outdoor site with roads and a vehicle, and a driven order between divisions. Works; competes with the game’s existing strength, which is 16 BALLPARKs of real arithmetic.

SPOT DEFER

Compartmentalised security is a genuine standing-instruction subject and it is the one part of this setting where a game should tread carefully. Defer on tone, not on fit.

LOB REJECT

An ordnance division makes this technically available and it should still be refused: nothing in this setting should be aimed at anything for fun.

SLATES Three appetites — the campaign is zero-sum

No conversion adds a stop. Each changes what an existing stop **does** while leaving what it **teaches** alone. Snapshot before, diff after: curriculumDelivery.mjs projecty --snapshot before.json , then --against before.json .

Minimum honest 3 STOPS · ONE SITTING				
32	d11	P	CHOICE → TALLY	Where the fragments go where the fragments go — counted, so $A = \lambda N$ is computed
41	d13	T	CHOICE → BALANCE	Where the energy in fission comes from where the energy comes from, closed as a ledger
43	d14	CM	CHOICE → SWEEP	Why separation is a physical problem separation as a curve rather than an explanation
CHOICE 8 → 5 (10%) · moves 11·36·2 → 8·37·4 · effective formats 5.5 → 6.0 · 2 new operate stops				

Balanced 6 STOPS · THE RECOMMENDATION

32	d11	P	CHOICE → TALLY	Where the fragments go
41	d13	T	CHOICE → BALANCE	Where the energy in fission comes from
43	d14	CM	CHOICE → SWEEP	Why separation is a physical problem
46	d15	T	CHOICE → CHAIN	Why light nuclei release energy too light nuclei, and the step that governs the release
14	d5	P	TRACE → PROBE	Are the neutrons missing, or the counts? inverse square, walked out from the source
36	d12	E	BALLPARK → HOLD	Critical path to an integrated trial pile power held on the rods

CHOICE **8 → 4** (8%) · moves **11·36·2 → 7·36·6** · effective formats **5.5 → 6.6** · **4** new operate stops, 1 of them arcade

Ambitious 9 STOPS · ADDS WORLD WORK

32	d11	P	CHOICE → TALLY	Where the fragments go
41	d13	T	CHOICE → BALANCE	Where the energy in fission comes from
43	d14	CM	CHOICE → SWEEP	Why separation is a physical problem
46	d15	T	CHOICE → CHAIN	Why light nuclei release energy too
14	d5	P	TRACE → PROBE	Are the neutrons missing, or the counts?
36	d12	E	BALLPARK → HOLD	Critical path to an integrated trial
20	d7	CM	DIAGNOSIS → TRIGGER	Phase diagrams a shutdown rule written before the number moves
27	d9	P	BALLPARK → CLOUD	High-speed imaging and radiography a yield estimate as a band the measurements permit
39	d13	E	SEQUENCE → TRIAL	Trinity field engineering the divisions, in the order the work actually goes

CHOICE **8 → 4** (8%) · moves **11·36·2 → 6·36·7** · effective formats **5.5 → 7.4** · **5** new operate stops, 2 of them arcade

THE ORDER OF WORK

1. Phase 0 — teach the coverage check to read an instrument's own board, and re-baseline curriculum-debt.json. Rewriting a stop that already teaches the thing is the exact failure the check exists to prevent. **2.** Snapshot. **3.** Edit the book, never the generated content — tools/import-book.mjs books/project-y.yml projecty --verify. **4.** npm run traps: every instrument's trap must fire. **5.** npm run check projecty and npm run drive projecty, right answer and wrong — ask what a player who understood nothing would score before believing any pass mark. **6.** --against the snapshot. A changed format is reported; a changed objective fails.

Counts computed from themes/projecty/content/ via engine/dev/curriculumDelivery.mjs; concept and equation lists authored in tools/syllabus.js. Format verdicts and slates are editorial. Run it with THEME=projecty npm run dev.